

MGTA 495: Healthcare Analytics (2 units)

SPRING 2026

CLASS TIME: Wednesday, 8:00 AM – 10:50 AM, April 1 – April 29, 2026.

LOCATION: WFH 4N128

PROFESSOR: Pooyan Kazemian

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OFFICE HOURS: Wednesday 11:00 AM – 12:00 PM in 3S147, or by appointment

TEACHING ASSISTANT: TBD

TA EMAIL: TBD

DESCRIPTION

This course provides a rigorous and practical introduction to healthcare decision modeling and analytics. We will start with cost-effectiveness analysis (CEA), which is the cornerstone of evidence-based resource allocation in healthcare. Students will learn how to evaluate competing health interventions by quantifying their costs and health outcomes using formal decision analysis methods. Building on this foundation, students will learn how to address uncertainty in decision models through deterministic and probabilistic sensitivity analyses. We will then focus on dynamic modeling frameworks widely used in health economics and policy analysis. Students will develop Markov cohort models and microsimulation (patient-level) models to represent chronic disease progression, competing risks, and patient heterogeneity, and analyze model outputs and visualize results. Through a combination of lectures, hands-on exercises, and real-world healthcare examples, students will gain the analytical and computational skills to build, analyze, and interpret decision models that inform health policy decisions.

PREREQUISITES

MGTA 453 or knowledge of probability and statistics.

OBJECTIVES

Upon completion of this course, students will be able to:

- Understand the fundamental principles of healthcare decision modeling and analytics.
- Construct and analyze decision models to evaluate the cost-effectiveness of healthcare interventions under uncertainty.
- Develop analytic solutions to practical health decision making problems for transforming data into knowledge.
- Apply deterministic and probabilistic sensitivity analysis techniques to assess the robustness of model outcomes.
- Develop and implement Markov cohort and microsimulation models to simulate disease progression, treatment effects, and patient heterogeneity.
- Integrate utility measurement methods into decision models and critically evaluate how outcome valuation choices influence cost-effectiveness results and policy recommendations.

CANVAS COURSE WEBSITE

To ensure that you receive course announcements in a timely manner, please update your notification settings so that Canvas Announcements and Inbox Conversations are forwarded to your email address.

SCHEDULE

Date	Class Topics	Due
Week 1	Introduction to Decision Analysis in Healthcare The Role of Decision Analysis in Healthcare; Overview of Methods for Healthcare Decision Making and Analytics Cost-Effective Analysis (CEA) Incremental Cost-Effectiveness Ratio (ICER); Strong vs. Weak Dominance; Comparison of Non-competing and Competing Interventions; Discounting; Net Monetary Benefit (NMB)	
Week 2	Managing Uncertainty in Healthcare Decision Models Types of Uncertainty in Healthcare Decision Models; Deterministic Sensitivity Analysis; Tornado Diagram; Probabilistic Sensitivity Analysis (PSA); Cost-effectiveness Acceptability Curve (CEAC); Building and Analyzing Healthcare Decision Models	Assignment 1 [C]
Week 3	Markov Models 1: Cohort Models When to Use Markov Models; Key Components of Markov Models; Markov Memoryless Assumption; Special Markov States; Markov Cohort Simulation; Markov Trace; Half-cycle Correction; Building and Analyzing Healthcare Markov Models	
Week 4	Markov Models 2: Microsimulation When to Use Microsimulation over Cohort Analysis; Estimating Transition Probabilities; Sensitivity Analysis with Microsimulation; First-Order vs. Second-Order Uncertainty; Building and Analyzing Healthcare Microsimulation Models	Assignment 2 [C]
Week 5	Valuing Health Outcomes Introduction to Utility Theory; Methods for Valuing Outcomes; Relationships Among Methods Course Project Cost-Effectiveness of Type 2 Diabetes Second-Line Medications	Course Project [G]

ASSIGNMENTS

All assignments will be posted on Canvas with a clear due date. Key:

- I – Independent, individual work only. No collaboration or consultation allowed.
- C – Collaboration with classmates is allowed. However, each student must submit a separately written up or produced deliverable (not a copy).
- G – Students may work together in groups and turn in one assignment for the entire group.

MATERIALS

The instructor will provide all required reading and lecture materials, including slides, data, references, and modeling exercises, which will be accessible through Canvas. The following textbooks have been used in developing the course content and are recommended for optional additional reading:

- Hunink, M. M., Weinstein, M. C., Wittenberg, E., Drummond, M. F., Pliskin, J. S., Wong, J. B., & Glasziou, P. P. (2014). *Decision Making in Health and Medicine: Integrating Evidence and Values*, 2nd Edition. Cambridge University Press.
- Drummond, M. F., Sculpher, M. J., Claxton, K., Stoddart, G. L., & Torrance, G. W. (2015). *Methods for the Economic Evaluation of Health Care Programs*. Oxford University Press.
- Briggs, A., Sculpher, M., & Claxton, K. (2006). *Decision Modelling for Health Economic Evaluation*. Oxford University Press.
- Neumann, P. J., Sanders, G. D., Russell, L. B., Siegel, J. E., & Ganiats, T. G. (2016). *Cost-effectiveness in Health and Medicine*. Oxford University Press.
- TreeAge Pro user manual and training materials.

GRADING

Exams and Deliverables	Percentage
Homework Assignments	40%
Class Attendance & Participation	20%
Course Project	40%
Total	100%

COURSE POLICIES

Course Format. The course will be offered in person and will consist of 5 weekly sessions. Sessions will include interactive lectures and hands-on modeling exercises. Students are encouraged to bring their laptops to follow along with lecture notes and coding exercises. All lectures will be recorded.

Attendance Policy. In-person attendance is mandatory for all students. If you must miss a class due to a valid reason (e.g., illness or family emergency), you must email the instructor with appropriate documentation (proof of valid reason) and watch the lecture recording in full within 7 days of the missed class to receive the attendance credit. Missing more than 2 classes will result in a failing grade ("F") for the course – no exceptions.

Expectations and Late Submissions. Students will be expected to participate in class discussions and complete two homework assignments, and a course project. All assignments must be turned in via Canvas by the specified deadline. Late submission will incur a penalty of 10% of the total points for each 24-hour period (or part thereof) past the deadline, and submissions are not accepted after solutions are posted on Canvas (usually 2-3 days after the due date). Students are encouraged to submit their assignments early if they anticipate delays closer to the due date.

Use of AI Tools. The use of AI tools such as ChatGPT, Gemini, Grok, and Claude is encouraged in this class. These tools have great potential to enhance your efficiency and improve the quality of your work. For example, you may use AI tools to brainstorm ideas, refine your writing, or help you conceptualize complex concepts. However, you should understand and explain every step of your analysis in your submission. You should never use AI tools to generate content that you simply present as your own, as doing so constitutes an academic integrity violation similar to submitting another person's work under

your name. You are also responsible for verifying the accuracy of any AI-generated content you use, since these tools can hallucinate or produce false information. In short, AI tools should serve as aids to your learning and creativity, not as substitutes for your own understanding and effort.

Changes. The instructors reserve the right to make changes during the term to any aspect of the syllabus that, in their judgment, are needed to achieve the learning objectives of the course.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.